

AN INTELLIGENT HYBRID SYSTEM FOR REAL-TIME WASTE CLASSIFICATION AND SEGMENTATION USING ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

A PROJECT – REPORT

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ABSTRACT

Waste management has become one of the most critical challenges in modern urban environments due to the increasing volume and complexity of waste generated daily. Inefficient segregation and reliance on manual processes often lead to environmental pollution, reduced recycling efficiency, and increased operational costs. To address these challenges, the proposed project introduces an Intelligent Hybrid System for Real-Time Waste Classification and Segmentation that leverages advanced artificial intelligence and machine learning techniques for automated waste analysis. The system primarily utilizes a state-of-the-art YOLOv8 object detection model to identify and localize multiple waste objects within an image, generating bounding boxes, class labels, and confidence scores in real time. To further enhance classification accuracy and contextual understanding, the system integrates the Google Gemini Vision API, which performs semantic analysis to determine material type, recyclability status, contamination level, and provides natural language disposal recommendations. This hybrid approach combines the speed of machine learning with the reasoning capability of generative AI, enabling more accurate and meaningful waste classification. The system is implemented through a Streamlit-based web application that allows users to capture or upload images and instantly visualize classification and segmentation results. Additionally, Explainable AI techniques using SHAP analysis are incorporated to identify the key visual features influencing model predictions, improving system transparency and reliability. The entire solution operates in a simulation-based environment, ensuring flexibility, scalability, and ease of deployment without requiring physical hardware. By integrating real-time detection, semantic analysis, and user-friendly interaction, the system provides an efficient and intelligent solution for automated waste management, contributing to improved recycling practices, reduced environmental impact, and the development of smart city infrastructure.

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LIST OF ABBREVIATIONS

S. NO	ABBREVIATION	EXPANSION
1	ML	Machine Learning
2	AI	Artificial Intelligence
3	YOLO	You Only Look Once
4	TFJS	TensorFlow.js
5	CNN	Convolutional Neural Network
6	UI	User Interface

CHAPTER 1

INTRODUCTION

Waste management has become one of the most pressing challenges in modern urban environments. The rapid growth of population, industrialization, and urbanization has significantly accelerated the volume of waste generated daily, placing enormous strain on existing waste collection, segregation, and disposal infrastructure. Traditional waste management approaches depend heavily on manual sorting processes and fixed collection schedules, which frequently result in inefficient segregation, overflowing bins, increased environmental contamination, and substantial operational costs. In most urban settings, waste workers are required to physically handle and sort mixed waste, exposing them to unhygienic and potentially hazardous conditions, while delivering inconsistent classification outcomes. To address these limitations, artificial intelligence and machine learning have emerged as powerful tools capable of automating and improving the accuracy of waste classification at scale. Recent advancements in computer vision, particularly object detection frameworks, have made it feasible to identify and classify waste items in real time using image data alone, without the need for physical sensors or manual intervention. These software-driven approaches offer a cost-effective, scalable, and intelligent alternative to traditional methods. In this project, we propose an Intelligent Hybrid System for Real-Time Waste Classification and Segmentation that combines two complementary AI technologies in a unified simulation framework.

The system integrates YOLOv8, a state-of-the-art object detection model, with the Google Gemini Vision API, a generative AI model capable of high-level semantic reasoning. Rather than operating these models sequentially, the proposed architecture runs them in parallel, significantly reducing inference latency while simultaneously improving classification depth and accuracy. The YOLOv8 model serves as the rapid detection layer, identifying waste objects within a captured image and generating bounding boxes, class labels, and confidence scores in a single forward pass. This output, along with the original captured frame, is concurrently processed by the Gemini-based pipeline, which performs semantic classification — determining material type, recyclability status, contamination level, and generating natural language disposal guidance.

Additionally, Gemini produces pixel-level segmentation masks for detected waste objects, enabling precise visual separation even in complex, overlapping waste scenes. The entire system is deployed as an interactive simulation through a Streamlit web application, which provides a real-time interface for live camera input, analysis output, and segmentation visualization. To further enhance system transparency, Explainable AI techniques using SHAP analysis are

incorporated, identifying which visual features such as material texture, object shape, and surface reflectance are most influential in recyclability predictions. The modular and scalable architecture of the proposed system ensures that it can be extended in future phases for IoT integration, edge deployment on devices such as Raspberry Pi, and smart city waste management applications.

1.1 PROBLEM STATEMENT

- Focuses on software implementation including dataset preparation, model training, and web deployment using YOLOv8 for real-time waste detection.
- Classifies waste into recyclable, non-recyclable, and organic categories with instant visual output (bounding boxes and labels).
- Designed with modular architecture to support future IoT hardware integration such as smart bins, sensors, and automated segregation.
- Scalable for additional deployment environments including mobile devices, cloud platforms, and municipal monitoring systems.
- Caters to various user groups like municipal authorities, institutions, corporate offices, and environmentally conscious individuals.
- Promotes responsible waste disposal by enabling accurate classification at the point of disposal.
- Contributes to environmental sustainability by improving recycling efficiency, reducing landfill waste, and lowering greenhouse gas emissions.
- Establishes a flexible, future-ready AI framework for smart waste management and smart city integration

1.2 OBJECTIVE OF THE PROJECT

The primary objective of this project is to design and develop an intelligent hybrid system for real-time waste classification and segmentation using advanced artificial intelligence and machine learning techniques. The project aims to build a fully software-based solution that automates the identification, categorization, and analysis of waste materials through image processing, eliminating the need for manual segregation and reducing dependency on hardware-based systems. A key objective is to integrate the YOLOv8 object detection model with the Google Gemini Vision API in a parallel inference architecture, enabling the system to achieve both high-speed detection and deep semantic understanding simultaneously. This hybrid approach is intended to improve classification accuracy, reduce processing latency, and provide

detailed insights such as material type, recyclability status, contamination level, and disposal recommendations.

Another important objective is to develop a real-time, user-friendly interface using Streamlit, which allows users to interact with the system through live camera input and instantly visualize classification and segmentation outputs. The project also aims to incorporate Explainable AI techniques, specifically SHAP analysis, to enhance transparency by identifying the key visual features influencing model predictions. This helps in overcoming the limitations of black-box AI systems and builds trust in automated decision-making. Furthermore, the system is designed with a modular and scalable architecture to ensure future compatibility with IoT devices, edge computing platforms, and smart city infrastructure. By focusing on accuracy, efficiency, interpretability, and scalability, the project aims to provide a practical and sustainable AI-driven solution that improves waste segregation, promotes recycling, and contributes to environmentally responsible urban development.

1.3 SCOPE OF THE PROJECT

The scope of this project encompasses the complete design, development, and simulation-based deployment of an AI-driven hybrid waste classification and segmentation system. The primary focus of this phase is on software simulation, wherein the entire system operates through a Python-based environment without the requirement of physical IoT hardware or embedded sensor infrastructure. This simulation-first approach enables rapid prototyping, iterative improvement, and comprehensive validation of the AI pipeline before any real-world hardware deployment is considered. The project covers multiple technical dimensions, including the integration of the YOLOv8 object detection model with the Google Gemini Vision API in a parallel inference architecture. This dual-model design is central to the scope of the work, as it addresses the limitations of single-model approaches by combining the speed of machine learning-based detection with the semantic reasoning capability of generative AI. The system is capable of classifying waste into categories including Plastic, Paper, Metal, Glass, Organic, and Mixed Waste, while also providing recyclability assessments and context-aware disposal recommendations.

The simulation environment is built entirely on Streamlit, which serves as the real-time web-based interface for user interaction, live camera feed processing, and visualization of detection and segmentation results. This removes the dependency on browser-based model execution or cloud server infrastructure, enabling a self-contained, locally deployable solution. The scope also includes the integration of Explainable AI through SHAP analysis, which

provides interpretable, feature-level insights into the system's classification decisions, addressing a critical gap in existing black-box AI waste management systems. Although this phase does not include physical hardware such as smart bins, ultrasonic sensors, or microcontrollers, the system architecture is deliberately designed to be hardware-ready.

The modular structure ensures that future phases can seamlessly integrate IoT sensors, edge computing devices, and cloud-based monitoring platforms without requiring significant rearchitecting of the core AI pipeline. This forward compatibility is a key aspect of the project scope, positioning the simulation as the intelligent core of a larger, eventually hardware-integrated waste management ecosystem. The system has direct applicability across a range of environments, including municipal waste stations, educational institutions, corporate campuses, and recycling facilities. By demonstrating that accurate, real-time, and explainable waste classification is achievable entirely through software simulation, this project establishes a practical and scalable foundation for AI-powered waste management aligned with smart city development goals and global environmental sustainability initiatives

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE SURVEY

2.1.1 Automated Waste Classification Using Deep Learning.

IEEE Xplore – Patel, R., & Sharma, K. (2022).

This paper presents an automated waste classification system based on deep learning models for intelligent waste segregation. The study utilizes Convolutional Neural Networks (CNNs) to classify waste images into biodegradable, recyclable, and non-recyclable categories. The system is trained on a large public dataset and achieves high accuracy in real-time testing. Results demonstrate that deep learning techniques can significantly reduce manual effort in waste sorting and support sustainable waste management through software-driven automation. In addition, the study emphasizes the importance of large-scale annotated datasets for improving classification performance. The authors highlight that model accuracy is highly dependent on data diversity and proper labeling. The system also demonstrates robustness under varying lighting conditions and backgrounds. However, the model requires significant computational resources, which may limit its deployment on low-power devices.

2.1.2 Machine Learning-Based Smart Waste Sorting System. Elsevier – Gupta, M., & Singh, T. (2021).

This research focuses on developing a machine learning model that automatically identifies different types of waste using image recognition. The system integrates supervised learning algorithms and image preprocessing techniques to enhance model performance. The software implementation provides an easy-to-use interface for uploading waste images and receiving instant predictions. The study highlights how ML-based classification can improve efficiency and accuracy in waste segregation compared to traditional manual methods. Furthermore, the research discusses the role of feature extraction techniques in improving model efficiency. It highlights that preprocessing steps such as normalization and resizing significantly enhance prediction accuracy. The system also supports scalability by allowing additional waste categories to be incorporated. However, performance may degrade when dealing with highly complex or mixed waste images.

2.1.3 Intelligent Waste Management Using CNN and Image Processing. Springer – Ahmad, Z.,

& Kumar, V. (2023).

The paper proposes an intelligent software framework that combines Convolutional Neural Networks and image processing techniques for smart waste classification. The model extracts image features to identify waste categories, achieving high precision with optimized training datasets. The system also integrates a web-based platform for user interaction, allowing real-time testing and data collection. Results prove the reliability of CNN architectures for automating waste management systems through software implementation. The study also explores optimization techniques to reduce training time while maintaining high accuracy. It emphasizes the importance of selecting appropriate hyperparameters for better model performance. Additionally, the integration of a web-based interface improves accessibility for users. Despite its advantages, the model struggles with real-time detection in highly cluttered environments. The model extracts image features to identify waste categories, achieving high precision with optimized.

2.1.4 Deep Learning for Waste Image Recognition and Segregation.

IEEE Transactions – Lee, J., & Park, H. (2020).

This study explores deep learning-based visual recognition for waste segregation tasks. The researchers implemented a transfer learning approach using pre-trained models such as VGG16 and ResNet50 to classify waste materials. The use of large-scale image datasets improved the classification accuracy and reduced training time. Experimental analysis shows that software-driven waste recognition systems can be integrated into digital platforms for smart city applications. Moreover, the research highlights the effectiveness of transfer learning in reducing training time and improving accuracy. The use of pre-trained models allows faster convergence and better generalization. The system also demonstrates adaptability across different datasets. However, the dependency on large datasets remains a limitation for achieving consistent performance.

2.1.5 Smart Waste Management Framework Using AI. Elsevier – Reddy, P., & Bansal, A. (2021).

This paper introduces an AI-driven waste management framework that utilizes machine learning and cloud computing for intelligent data processing. The system predicts waste categories based on image input and stores results in a cloud database for further analysis. The software focuses on scalability, real-time classification, and efficient data storage. The study demonstrates how AI

models can support municipal waste systems by enabling intelligent, software-based decision-making. The framework also focuses on integrating cloud-based storage for efficient data handling and analysis. It enables continuous monitoring and improvement of classification models using stored data. The study highlights the importance of real-time processing in urban waste management systems. However, reliance on cloud infrastructure may introduce latency and connectivity issues.

2.1.6 Waste Classification Using Transfer Learning Models. Springer – Li, C., & Wang, Y. (2022).

The research employs transfer learning techniques such as MobileNet and InceptionV3 to improve waste image classification accuracy. The system is implemented as a web application where users can upload images for instant classification. The model performs exceptionally well on heterogeneous datasets with minimal computational cost. The paper emphasizes the adaptability of software-based ML models for large-scale waste management solutions. Additionally, the research demonstrates that lightweight models can achieve high accuracy with reduced computational cost. It highlights the importance of model selection based on deployment requirements. The system also supports real-time classification through web-based platforms. However, performance may vary depending on the complexity of the dataset used.

2.1.7 Data-Driven Waste Segregation System Using Artificial Intelligence.

IEEE Access – Nair, P., & Das, R. (2023).

This paper proposes a data-driven waste management system that leverages artificial intelligence to categorize waste images. The system integrates a CNN-based model with a backend database to record and analyze waste segregation patterns. Experimental evaluation demonstrates consistent classification accuracy and highlights the benefits of using AI-powered software tools in environmental sustainability initiatives. The study further emphasizes the role of data analytics in improving waste management efficiency. By storing classification results, the system enables pattern analysis and predictive insights. This helps in optimizing waste collection strategies in urban areas. However, maintaining large datasets may require significant storage and processing capabilities.

2.1.8 Software-Based Waste Detection Using Computer Vision.

Elsevier – Khan, S., & Mehta, D. (2021).

The study presents a software framework that uses computer vision algorithms for waste detection and classification. The proposed method applies image segmentation, feature extraction, and ML-based classification to differentiate between recyclable and non-recyclable materials. The software operates in real time and supports integration with web applications. The results confirm that computer vision combined with ML provides an efficient solution for intelligent waste monitoring. The research also highlights the importance of image segmentation techniques in improving classification accuracy. It demonstrates that combining feature extraction with machine learning enhances system performance. The system is capable of real-time detection under controlled environments. However, accuracy may decrease when handling overlapping or unclear waste objects.

2.1.9 Sustainable Waste Sorting through AI and Cloud Integration. Springer – Rao, L., & Verma, P. (2022).

This research integrates AI-based waste classification with cloud storage for centralized data management. The software processes image data locally using CNN models and uploads classification results to the cloud for tracking and analytics. The framework ensures scalability and continuous learning from new data inputs. The study proves that cloud-based AI systems enhance efficiency, reduce manual workload, and support sustainable waste management operations. Furthermore, the study focuses on scalability and adaptability of cloud-integrated systems. It enables centralized monitoring and efficient resource management across multiple locations. The framework supports continuous learning through data updates. However, dependency on internet connectivity may limit its usability in remote areas.

2.1.10 Smart Waste Identification Using Image-Based Machine Learning.

IEEE Xplore – Ahmed, F., & Patel, S. (2023).

This paper introduces an image-based waste identification system using machine learning algorithms implemented in a web environment. The proposed model uses supervised classification techniques to distinguish waste categories and improve sorting accuracy. The web application allows real-time user interaction and provides immediate classification feedback. The study highlights the effectiveness of integrating ML models with user-centric software design to promote cleaner and smarter cities. The research also discusses user interaction as a key factor in system usability. The web-based interface simplifies the process of waste classification for non-technical users. It highlights the importance of real-time feedback in improving user engagement.

However, the system may face challenges in maintaining accuracy for diverse and unstructured waste inputs.

2.2 SURVEY ON PROJECT

The survey on the Smart Waste Management System using Machine Learning focuses on understanding the role of intelligent software solutions in improving waste segregation efficiency. Traditional waste management systems rely heavily on manual sorting, which is time-consuming, labor-intensive, and prone to human error. With the increasing volume of waste generated in urban areas, there is a growing need for an automated and scalable software-driven solution. Through the integration of machine learning algorithms, particularly Convolutional Neural Networks (CNNs), waste classification can be automated with higher accuracy and reliability.

The review of existing works highlights that most earlier systems used IoT-based smart bins and hardware sensors for waste detection. While these systems contributed to better monitoring, they lacked the intelligence to classify waste effectively. Recent studies indicate a significant shift toward software-based approaches that rely on computer vision and machine learning to analyze waste images. CNN models such as VGG16, ResNet, and MobileNet have been successfully implemented in similar projects for object recognition and waste categorization. These models process image datasets containing various waste types and learn patterns that distinguish between biodegradable, non-biodegradable, and recyclable materials.

Furthermore, surveys indicate that integrating a trained ML model into a web or mobile application allows users to interact with the system easily by uploading images for instant classification. This makes the solution more accessible, scalable, and suitable for deployment in smart city environments. Various research papers emphasize that using a centralized software platform helps collect and store data for continuous model improvement and analytics-based decision-making. The availability of open-source datasets, such as TrashNet and WasteNet, has also contributed to developing more accurate and efficient models.

In the context of the proposed project, the survey confirms that software-focused waste classification systems provide a cost-effective and sustainable alternative to hardware-intensive models. They reduce dependency on sensors, minimize maintenance costs, and allow rapid deployment across different regions. With the advancement in deep learning frameworks like TensorFlow and PyTorch, implementing and fine-tuning classification models has become more efficient. foundation for the proposed project, which aims to combine IoT-based crash detection

with real-time alert mechanisms to ensure timely emergency response.

2.2.1 EXISTING SYSTEM

The existing waste management systems primarily rely on manual segregation methods or IoT-based sensor mechanisms for waste detection. In manual systems, workers sort the waste into categories such as biodegradable, non-biodegradable, and recyclable, which is inefficient, time-consuming, and prone to human error. These traditional methods also expose workers to unhygienic conditions, increasing health risks and reducing overall productivity. Some smart city initiatives have introduced IoT-enabled bins equipped with ultrasonic sensors to detect waste levels and alert authorities when bins are full. While these systems help monitor waste collection, they do not classify waste or provide insights into its composition. In other words, they focus on waste quantity rather than waste type. Additionally, these hardware-based systems are costly to deploy and maintain, making them less scalable for large cities or developing regions. From a software standpoint, existing applications are limited to data monitoring and reporting, without incorporating machine learning or computer vision for intelligent waste classification. The lack of automation and intelligent analysis makes these systems inefficient in promoting effective recycling and waste segregation. Therefore, there is a need for a fully software-based intelligent system that leverages image processing and machine learning to automate classification and support sustainable waste management practices. Additionally, these hardware-based systems are costly to deploy and maintain, making them less scalable for large cities or developing regions. From a software standpoint, existing applications are limited to data monitoring and reporting, without incorporating machine learning or computer vision for intelligent waste classification. The lack of automation and intelligent analysis makes these systems inefficient in promoting effective recycling and waste segregation.

The review of existing works highlights that most earlier systems used IoT-based smart bins and hardware sensors for waste detection. While these systems contributed to better monitoring, they lacked the intelligence to classify waste effectively. Recent studies indicate a significant shift toward software-based approaches that rely on computer vision and machine learning to analyze waste images. Traditional waste management systems rely heavily on manual sorting, which is time-consuming, labor-intensive, and prone to human error. With the increasing volume of waste generated in urban areas, there is a growing need for an automated and scalable software-driven solution and suitable for deployment in smart city environments.

The existing waste management systems primarily rely on manual segregation methods or IoT-based sensor mechanisms for waste detection. In manual systems, workers sort the waste into categories such as biodegradable, non-biodegradable, and recyclable, which is inefficient, time-consuming, and prone to human error. These traditional methods also expose workers to unhygienic conditions, increasing health risks and reducing overall productivity. Some smart city initiatives have introduced IoT-enabled bins equipped with ultrasonic sensors to detect waste levels and alert authorities when bins are full. While these systems help monitor waste collection, they do not classify waste or provide insights into its composition. In other words, they focus on waste quantity rather than waste type. Additionally, these hardware-based systems are costly to deploy and maintain, making them less scalable for large cities or developing regions.

From a software standpoint, existing applications are limited to data monitoring and reporting, without incorporating machine learning or computer vision for intelligent waste classification. The lack of automation and intelligent analysis makes these systems inefficient in promoting effective recycling and waste segregation. Therefore, there is a need for a fully software-based intelligent system that leverages image processing and machine learning to automate classification and support sustainable waste management practices.

2.2.2 DRAWBACKS OF EXISTING SYSTEM

- Existing systems rely heavily on manual segregation, which is time-consuming and often leads to inaccurate sorting of waste.
- Lack of automation prevents efficient classification and slows down the overall waste management process.
- Most systems use IoT sensors only for waste level detection, without integrating any machine learning or intelligent classification mechanism.
- High operational and maintenance costs arise due to dependency on hardware components and physical infrastructure.
- Absence of image-based recognition limits the system's ability to identify waste types effectively.
- Data handling and tracking are inefficient, as existing systems lack centralized databases for analysis and decision-making.

- Low scalability restricts widespread deployment across cities because of the high setup and maintenance requirements.
- Inconsistent accuracy and reliability in waste sorting reduce the effectiveness of recycling and environmental sustainability efforts.

2.2.3 PROPOSED SYSTEM

The proposed system focuses on developing an intelligent software-based waste management platform that utilizes machine learning and image processing to automate the segregation of waste. It aims to overcome the inefficiencies of manual and hardware-based systems by implementing a Convolutional Neural Network (CNN) model trained on large datasets of waste images. The model is capable of analyzing visual features and accurately classifying waste into categories such as biodegradable, non-biodegradable, and recyclable. This approach eliminates the need for physical sensors or human intervention, providing a fully automated and efficient classification process. The system also includes a web-based interface, allowing users to easily upload images of waste items. Once uploaded, the trained ML model processes these images and generates instant, accurate classification results. The software incorporates essential components such as data preprocessing, model training, validation, and optimization, ensuring high accuracy and performance. Additionally, the system maintains a database to store image data, results, and analytics for continuous model improvement and decision-making. This enables scalability and adaptability, allowing the system to be expanded for different waste categories or integrated into larger smart city platforms. By emphasizing the software dimension, the proposed system minimizes hardware costs, simplifies maintenance, and ensures faster deployment. Overall, this intelligent, software-driven solution promotes efficient waste segregation, sustainability, and environmental awareness, aligning with modern technological initiatives toward smart and eco-friendly urban development.

2.2.4 ADVANTAGES OF PROPOSED SYSTEM

- **Automated Waste Classification:** The system uses a trained machine learning model to automatically identify and categorize waste, eliminating the need for manual segregation.
- **High Accuracy and Efficiency:** By leveraging image recognition through CNN models, the system ensures faster and more precise classification compared to traditional methods.

- **Software-Centric Approach:** The solution operates entirely through software, reducing hardware dependency and minimizing installation and maintenance costs.
- **Real-Time Processing:** The web-based interface enables users to upload waste images and receive instant classification results, making the process seamless and efficient.
- **Scalability and Flexibility:** The modular software design allows easy integration of additional waste categories or connection with IoT devices in future enhancements.
- **Data Storage and Analytics:** The system can maintain a centralized database for storing images and classification results, supporting performance tracking and model improvement.
- **Cost-Effective Solution:** Since it does not rely on expensive sensors or physical components, the system provides a low-cost alternative for smart waste management.
- **User-Friendly Interface:** The application offers an intuitive and simple interface that can be accessed by anyone, including municipal workers and the general public.
- **Environmentally Sustainable:** By improving the accuracy of waste segregation, the system promotes recycling and proper disposal, contributing to cleaner and greener surroundings.
- **Supports Smart City Development:** The software aligns with modern smart city initiatives, showcasing how artificial intelligence can be used to enhance environmental management and urban sustainability.

CHAPTER 3

SYSTEM SPECIFICATIONS

3.1 SOFTWARE REQUIREMENTS

- Operating System : Windows 10/Linux/macOS
- Programming Languages : Python, JavaScript
- Object Detection Framework : Ultralytics YOLOv8
- Generative AI API : Google Gemini Vision API
- Frontend Framework : Streamlit
- Libraries : NumPy, OpenCV, Pillow, Matplotlib
- Explainability Tool : SHAP (Shapely Additive Explanations)
- Development Environment : VScode/ Jupyter Notebook
- Package Manager : Pip/ Virtualenv
- Version Control : Git/ Github

3.2 HARDWARE REQUIREMENTS

- Processor : Intel i5 / Ryzen 5 or above
- RAM : Minimum 8 GB
- Storage : 10 GB free space
- GPU (Interface) : NVIDIA GPU (optional) / Google Colab Free GPU
- Camera : Standard Laptop Or USB webcam
- Internet Connection : Required for Google Gemini API

3.3 TECHNOLOGY USED

The Smart Waste Management System is developed using a combination of advanced software technologies, machine learning frameworks, and modern web-based tools. These technologies work together to enable efficient data processing, model training, real-time image classification, and user interaction through a seamless interface. The selection of technologies ensures scalability, flexibility, and high performance while maintaining a software-centric architecture.

3.3.1 DEVELOPMENT PLATFORMS AND FRAMEWORKS

The system is developed primarily using Python, a versatile and widely adopted programming language for artificial intelligence and data-driven applications. Python provides extensive libraries and community support, making it ideal for building machine learning models and integrating them into a web application. The main framework used in the project is Streamlit, a modern, open-source Python framework that simplifies the creation of interactive web applications for data science and machine learning. Streamlit allows developers to create dashboards and interfaces directly from Python scripts without requiring HTML, CSS, or JavaScript. This framework supports real-time interaction, file uploads, and immediate display of model predictions, making it a perfect choice for this project's software-centric design. In addition, TensorFlow and Keras serve as key frameworks for building and training the Convolutional Neural Network (CNN) model used for waste classification. These frameworks simplify deep learning model design through high-level APIs and GPU acceleration support. The overall integration of Streamlit and TensorFlow ensures a smooth flow between model execution and user interaction. The combination of these platforms and frameworks provides a seamless, efficient, and scalable foundation for building, training, and deploying the Smart Waste Management system within a single software environment.

The system is primarily developed using Python, which is widely recognized for its simplicity, readability, and extensive support for artificial intelligence and data science applications. Python provides a vast ecosystem of libraries and frameworks that simplify complex tasks such as image processing, model training, and deployment.

3.3.2 MACHINE LEARNING AND AI TOOLS

The core functionality of this project relies on machine learning and artificial intelligence for automatic waste classification. The Convolutional Neural Network (CNN) algorithm forms the backbone of the system, enabling accurate detection and categorization of waste images into

biodegradable, non-biodegradable, and recyclable types. TensorFlow is used as the main deep learning framework due to its flexibility, efficiency, and compatibility with GPU computation. It provides various layers, optimizers, and pre-trained models that make training and fine-tuning easier. Alongside TensorFlow, Keras acts as a user-friendly interface for model design, training, and evaluation. NumPy and Pandas are used for data manipulation and preprocessing, while Matplotlib helps visualize model accuracy and loss during training. For handling image data, OpenCV and Pillow libraries are employed to resize, normalize, and prepare images for CNN input. These tools collectively enable efficient model training, testing, and real-time prediction, forming the AI foundation that powers the intelligent waste classification process in this software system. The intelligence of the system is powered by advanced machine learning and computer vision techniques. The project utilizes YOLOv8 (You Only Look Once version 8), a state-of-the-art object detection algorithm known for its high speed and accuracy. YOLO processes the entire image in a single pass, enabling real-time detection and classification of multiple waste objects with bounding boxes and confidence scores. In addition to YOLOv8, the system integrates the Google Gemini Vision API, which enhances the model's capabilities by providing semantic understanding of waste materials. Unlike traditional models, Gemini can interpret context, identify material composition, and generate meaningful insights such as recyclability status and disposal instructions.

3.3.3 DATABASE AND STORAGE

The project includes a lightweight and efficient storage solution designed to manage images, predictions, and user interactions. For simplicity and ease of integration, SQLite is used as the local database. SQLite offers a self-contained, serverless database engine that fits perfectly for small-to-medium-scale software projects. It stores essential information such as image metadata, prediction results, and timestamps for model evaluation. In addition to SQLite, the project also uses local file storage to handle uploaded images during runtime. This allows the Streamlit interface to process images directly without external dependencies. If the system is scaled for larger deployments, the database can easily be upgraded to MySQL or MongoDB to support cloud-based storage and multiple user access. The integration of database and storage components ensures that the software can maintain a structured record of classified waste data, which can be used later for analysis, system improvement, or integration with city-wide waste monitoring dashboards. For handling images during runtime, the system uses local file storage, which enables quick access and processing without relying on external servers. This ensures faster performance and reduces system dependency. The architecture is designed to be scalable, allowing

future integration with advanced databases such as MySQL, PostgreSQL, or MongoDB for cloud-based storage and large-scale data management. This flexibility supports expansion into enterprise-level or smart city applications.

3.3.4 DEVELOPMENT ENVIRONMENT

The system is developed using the Python programming environment on Visual Studio Code (VS Code), a powerful and lightweight integrated development environment (IDE). VS Code provides features such as syntax highlighting, debugging tools, and integrated terminal support that make coding and testing more efficient. The model training process is carried out in Jupyter Notebook, which provides an interactive environment for experimenting with data preprocessing, visualization, and neural network training. The final web application is implemented and run using the Streamlit environment, which simplifies deployment and testing through direct command execution (`streamlit run app.py`). The project dependencies and versions are managed using pip or virtual environments (venv) to ensure smooth installation and portability across systems. This setup ensures that the system is easy to develop, maintain, and deploy on various operating systems such as Windows or Linux. Dependency management is handled using pip and virtual environments (venv), ensuring that all required libraries are properly installed and isolated. This improves reproducibility and avoids compatibility issues across different systems.

3.3.5 API AND SERVICES

In this project, a separate REST API is not required, as Streamlit itself handles both the user interface and backend logic. Streamlit internally serves as the application layer where users interact with the trained machine learning model directly through the web interface. However, in an extended version of the system, APIs can be integrated for scalability — for example, connecting the waste classification model to IoT devices, mobile apps, or cloud platforms. Such APIs could be built using Flask or FastAPI, enabling real-time communication between the model server and external systems. Currently, Streamlit manages the core functionality, image upload, and prediction handling within one framework, ensuring fast processing and smooth integration without external API dependencies. The system integrates the Google Gemini Vision API, which plays a crucial role in providing advanced semantic analysis of waste images. This API enables the system to go beyond basic classification by offering contextual understanding, segmentation, and natural language insights. Currently, no external REST API is required, as Streamlit handles both frontend and backend operations internally. However, the system is designed to support future API integration using frameworks like Flask or FastAPI, which can enable communication

with mobile applications, IoT devices, or cloud platforms. This flexibility allows the system to evolve into a distributed architecture where multiple services interact in real time.

3.3.6 MODEL SIMULATION ENVIRONMENT & DEPLOYMENT

The simulation environment for this project is built entirely on Python, providing a self-contained, locally executable system without dependency on external servers or embedded hardware. The core simulation is powered by Streamlit, which serves as both the frontend interface and application runtime, handling real-time camera input, model inference, and result visualization within a single unified environment. The system is launched using the command `streamlit run app.py`, which deploys the web interface directly in the browser without requiring separate server configuration. All inference operations, including YOLOv8 detection and Gemini API processing, run within the same Python runtime, ensuring seamless data flow between components. Virtual environments managed through pip and venv isolate project dependencies, ensuring reproducibility across Windows, Linux, and macOS systems. For version control, Git is used to track changes and the repository is maintained on GitHub for backup and organized development. The entire system operates in a software-based simulation environment, eliminating the need for physical hardware components. The simulation is powered by Python and executed locally, ensuring portability and ease of deployment. The system follows a parallel inference architecture, where YOLOv8 and Gemini API process input images simultaneously. This significantly reduces processing time and improves efficiency compared to sequential execution. The deployment is handled through Streamlit, which provides a browser-based interface without requiring additional server configuration. The system can run on multiple operating systems, including Windows, Linux, and macOS. Version control is managed using Git, and the project repository is maintained on GitHub, enabling collaborative development, tracking changes, and maintaining backups.

3.3.7 TESTING AND DEBUGGING TOOLS

Testing and debugging play a crucial role in ensuring that the software functions reliably. The project uses PyTest and unittest frameworks in Python to test individual modules, ensuring that image uploads, preprocessing, and predictions work as expected. For debugging and error tracking, VS Code Debugger and Streamlit's built-in console logs are used to identify and resolve runtime issues quickly. Model performance is evaluated using confusion matrices, accuracy, and loss curves, ensuring that predictions are consistent and accurate. Additionally, during model training and validation, TensorBoard is used to visualize metrics and detect overfitting or

underperformance. This combination of testing and debugging tools ensures that the application is stable, efficient, and ready for real-world use. To ensure reliability and performance, the system uses various testing and debugging tools. PyTest and unittest frameworks are used for validating individual modules and ensuring correct functionality. Debugging is performed using the VS Code Debugger and Streamlit console logs, which help identify runtime errors and optimize performance. For monitoring model training, TensorBoard is used to visualize metrics such as loss, accuracy, and learning curves. Performance evaluation is conducted using metrics like Precision, Recall, F1 Score, and Mean Average Precision (mAP). These metrics ensure that the model delivers accurate and consistent predictions. Overall, the combination of these tools ensures that the system is robust, efficient, and ready for real-world deployment.

CHAPTER 4

SYSTEM DESIGN

4.1 ER DIAGRAM

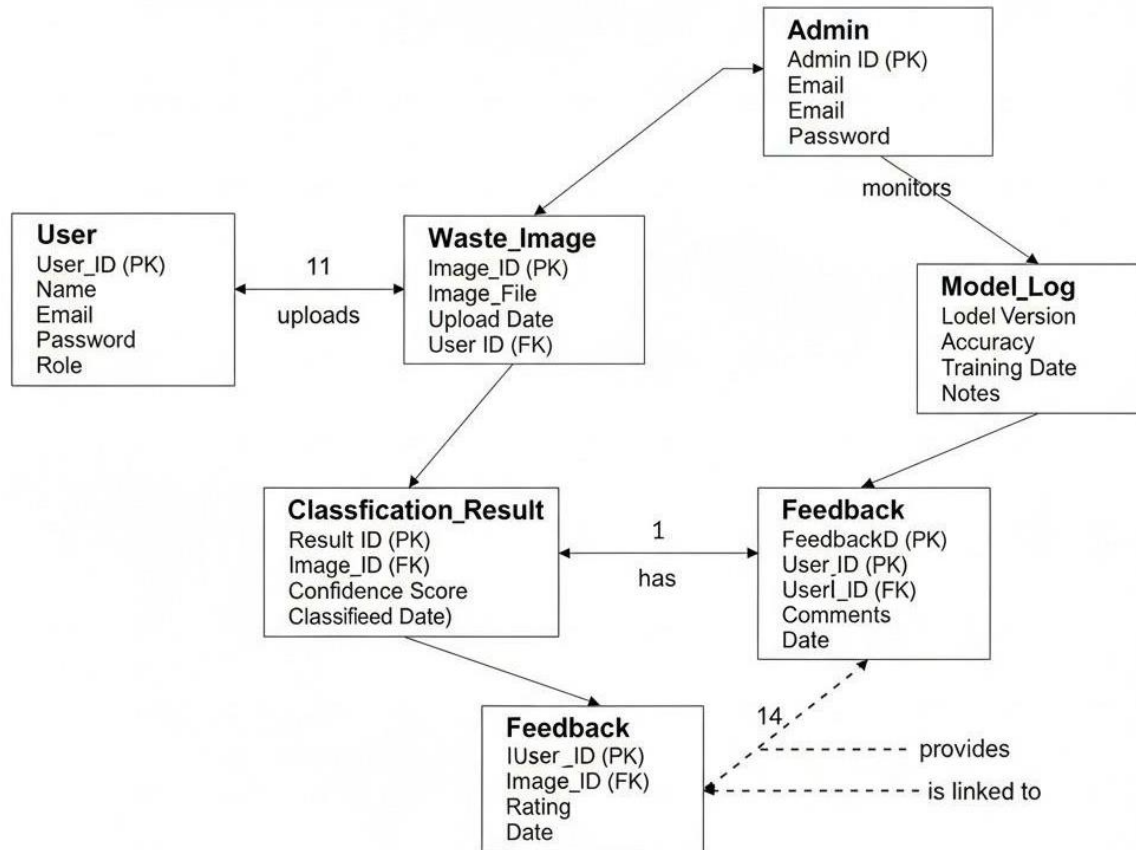


Fig 4.1.1 ER DIAGRAM

The ER diagram illustrates the key entities and relationships within the smart waste management system. The Entity-Relationship (ER) diagram represents the logical structure of the Smart Waste Management System by illustrating the key entities, their attributes, and the relationships between them. The primary entities in the system include User, Waste Image, Classification Result, and System Output. Each user interacts with the system by uploading or capturing waste images through the interface, which are then stored along with relevant metadata such as timestamp, image ID, and user details. The Waste Image entity is directly linked to the Classification Result entity, where the machine learning model processes the input image and generates outputs such as waste category, confidence score, and segmentation details. Additionally, the System Output entity maintains processed results including recyclability status, material composition, and disposal recommendations. The relationships defined in the ER diagram ensure proper data flow and integrity, allowing efficient storage, retrieval, and analysis of

waste classification data. This structured representation helps in designing a scalable database system that supports future enhancements such as cloud integration, user analytics, and large-scale smart city deployments.

4.2 DATA FLOW DIAGRAM

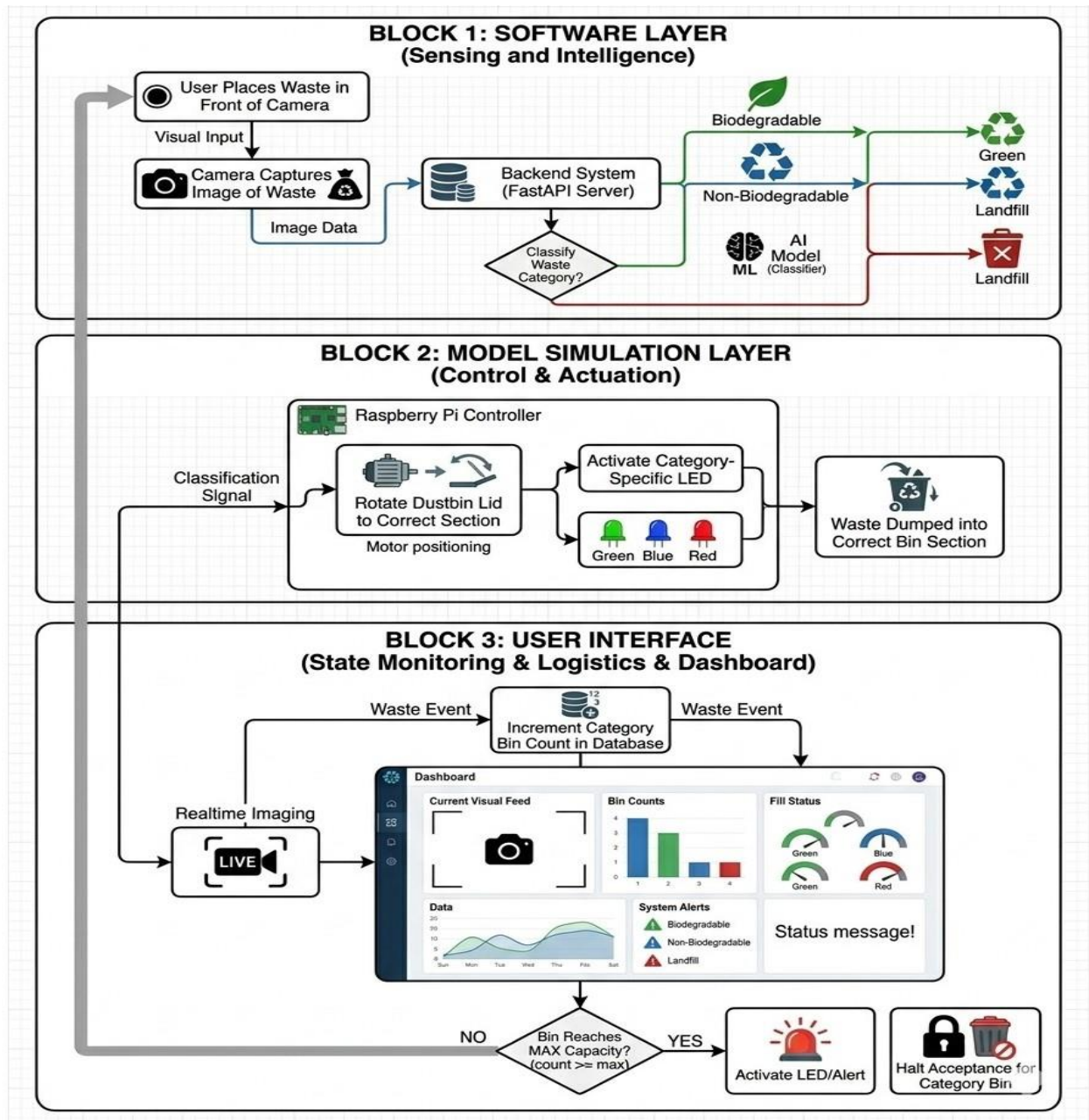


Fig 4.2.1 DATA FLOW DIAGRAM

This diagram illustrates the complete workflow of the proposed Smart Waste Detection System, starting from dataset acquisition and preprocessing to YOLOv8-based model training and real-time detection. The pipeline integrates dataset conversion, performance evaluation, and deployment of the trained TensorFlow.js model on a web interface, enabling users to receive instant waste categorization results through bounding boxes and confidence scores. The Data Flow Diagram (DFD) illustrates the end-to-end flow of data within the Smart Waste Management System, highlighting how input data is processed and transformed into meaningful outputs. The process begins with the user capturing or uploading an image of waste through the Streamlit-based interface. This image is then passed to the preprocessing stage, where operations such as resizing, normalization, and format conversion are performed to prepare the data for model input. The processed image is simultaneously sent to the YOLOv8 object detection model and the Gemini Vision API in a parallel inference pipeline. YOLOv8 identifies and localizes waste objects by generating bounding boxes and class labels, while the Gemini API performs semantic analysis, providing insights such as material type, recyclability status, and contamination level. The outputs from both models are then combined and sent to the visualization module, where results are displayed to the user in real time. The system may also store classification results in a database for further analysis and performance tracking. This structured data flow ensures efficient processing, reduced latency, and accurate real-time waste classification.

4.3 SYSTEM ARCHITECTURE

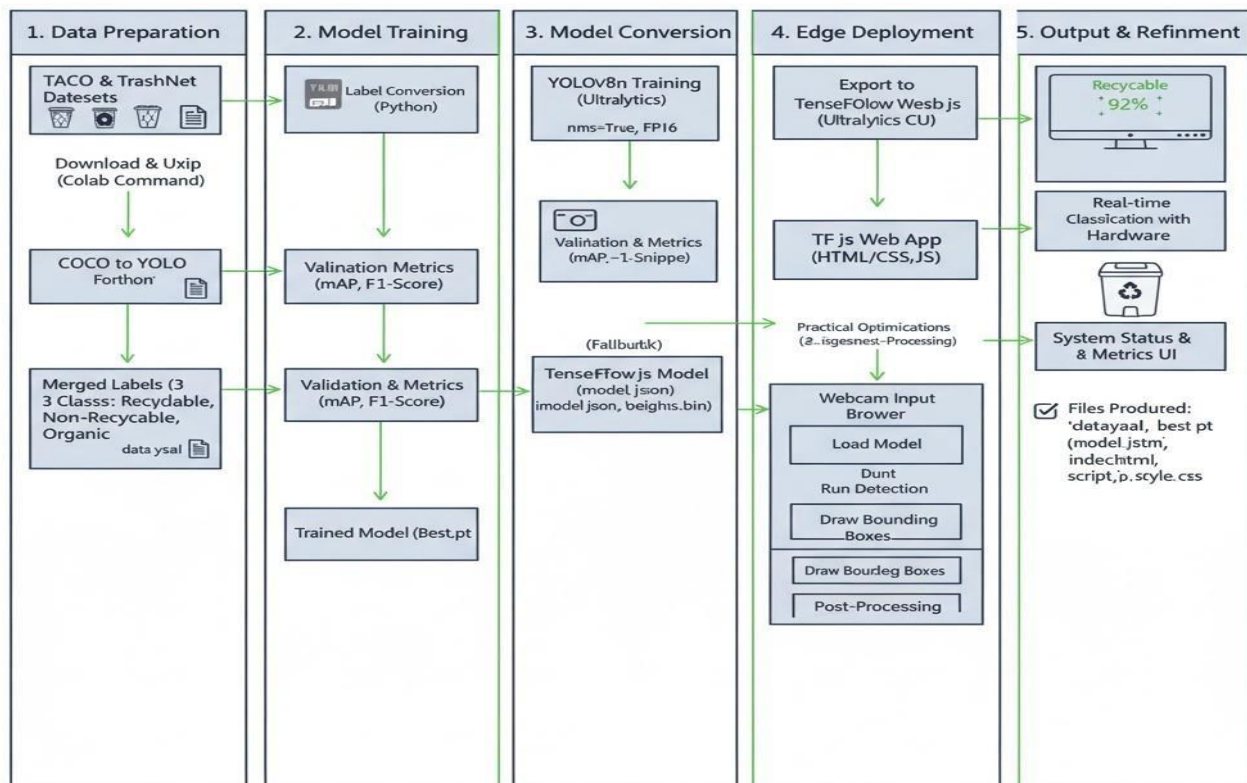


Fig 4.3.1 SYSTEM ARCHITECTURE

The figure 4.3.1 represents the complete development and deployment pipeline of the Smart Waste Detection System. It begins with dataset preprocessing and label merging, followed by YOLOv8 model training and performance evaluation. The trained model is then converted into TensorFlow.js format for web-based execution. Finally, real-time object detection is achieved through a browser interface, providing classification results with system performance metrics for improved waste management. The system architecture diagram presents the overall design and interaction between different components of the Smart Waste Management System. The architecture follows a modular and layered approach, consisting of data acquisition, processing, model inference, and user interface layers. Initially, input data is collected through a camera or uploaded image, which serves as the primary input to the system. This data is then passed through the preprocessing layer, where it is standardized for compatibility with machine learning models. The core of the architecture lies in the parallel inference mechanism, where the YOLOv8 model and the Gemini Vision API operate simultaneously to perform object detection and semantic analysis. The YOLOv8 model focuses on fast and accurate detection of waste items, while the Gemini API enhances the output by providing contextual understanding and segmentation. The results from both models are integrated and forwarded to the visualization layer, which is implemented using Streamlit to display outputs such as bounding boxes, classification labels, and recyclability information. The architecture is designed to be scalable and flexible, allowing future integration with IoT devices, cloud platforms, and edge computing systems. This modular design ensures efficient processing, easy maintenance, and adaptability for real-world smart waste management applications.

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 MODULE

The Smart Waste Management system is divided into several modules that work together to ensure efficient waste classification and system integration. Each module handles a specific stage, from collecting and preparing image data to training, evaluating, and deploying the machine learning model through a web interface. The modular approach allows for flexibility, scalability, and easy maintenance, ensuring that the software functions effectively and can be upgraded as technology evolves.

5.2 MODULES DESCRIPTION

Each module in the system performs a specific role in building an intelligent, software-based waste management solution. From preparing the dataset to deploying the trained model, each stage contributes to improving efficiency and automation. The detailed description of individual modules below explains the internal workflow, logic, and the software tools used to achieve the project's objectives.

5.2.1 DATASET PREPARATION MODULE

This module is responsible for collecting, organizing, and labeling waste images used for model training. It ensures that the data is clean, diverse, and representative of real-world waste scenarios. Import images from public datasets like TACO (Trash Annotations in Context). Perform data cleaning, removing duplicates or corrupted files. Apply annotation mapping, grouping items into three major categories: Recyclable, Non Recyclable, and Organic. Split the dataset into training, validation, and testing subsets (typically 70:20:10). Perform data augmentation (rotation, flipping, brightness adjustment) to improve model generalization. In addition to basic data collection, careful attention is given to ensuring dataset diversity to improve the robustness of the model. Images are selected from multiple sources to include variations in lighting conditions, backgrounds, and object orientations, which helps the model generalize better in real-world scenarios. The preprocessing stage includes resizing images to a fixed dimension compatible with YOLOv8 input requirements, along with normalization to scale pixel values for efficient training. Label consistency is maintained by mapping similar categories into unified classes to avoid ambiguity during training. Data augmentation techniques such as rotation, horizontal flipping, zooming, and brightness adjustment

are applied to artificially increase dataset size and reduce overfitting. The dataset is also validated to remove noisy or mislabeled samples that could negatively impact model accuracy. Proper directory structuring is followed to organize training and validation datasets, ensuring smooth integration with the training pipeline. This comprehensive dataset preparation process plays a critical role in improving the overall performance and reliability of the system.

5.2.2 MODEL DEVELOPMENT MODULE

The heart of the system lies in its ability to identify and classify waste items automatically. This is achieved through object detection, a computer vision technique that locates and labels objects within an image. For this project, the YOLOv8 (You Only Look Once version 8) algorithm is adopted due to its exceptional balance between speed and accuracy. Unlike traditional image classifiers that only identify a single class per image, YOLO detects multiple objects simultaneously and provides bounding boxes with confidence scores — essential for detecting multiple waste items in one frame. Among YOLOv8 variants (Nano, Small, Medium, Large), YOLOv8n (Nano) was chosen for this phase as it provides the ideal trade-off between computational efficiency and accuracy.

The model development phase focuses on configuring and optimizing the YOLOv8 architecture to suit the waste classification problem. The model is initialized using pre-trained weights, which allows it to leverage previously learned features from large datasets, thereby reducing training time. Customization is performed by modifying the output layers to match the specific waste categories defined in the dataset. Anchor boxes are automatically adjusted by the YOLO framework to better detect objects of varying sizes present in waste images. The model is configured to process images in real time, ensuring a balance between detection speed and accuracy. Hyperparameters such as learning rate, batch size, and number of epochs are carefully tuned to achieve optimal performance. The development process also includes testing different YOLOv8 variants to select the most efficient model for deployment. Additionally, techniques such as dropout and regularization are considered to prevent overfitting.

This phase ensures that the model is well-structured, optimized, and ready for effective training and deployment. Customization is performed by modifying the output layers to match the specific waste categories defined in the dataset. Anchor boxes are automatically adjusted by the YOLO framework to better detect objects of varying sizes present in waste images. The model is configured to process images in real time, ensuring a balance between detection speed and accuracy.

5.2.3 MODEL TRAINING AND EVALUATION MODULE

Training was conducted entirely on Google Colab using free GPU support. The environment included all dependencies such as PyTorch, Ultralytics YOLO, NumPy, and OpenCV. The dataset was uploaded to the Colab workspace, and the data.yaml file was configured with image directories and class labels.

Training Process

1. **Loading the Pretrained Model:** The base YOLOv8n model, pre-trained on the COCO dataset, was imported for transfer learning.
2. **Fine-tuning on Custom Data:** The model's final layers were retrained on the custom waste dataset.
3. **Monitoring Training:** Real-time loss graphs and accuracy metrics were displayed in Colab.
4. **Validation Testing:** After each epoch, the model was evaluated on the validation set to ensure that it was not overfitting.

During training, the model progressively learned to distinguish between waste categories based on shape, color, and texture patterns.

Evaluation Metrics: The model's performance was assessed using the following metrics:

- Precision:** Measures the proportion of correctly predicted waste items among all predictions.
- Recall:** Measures the proportion of actual waste items detected by the model.
- mAP (Mean Average Precision):** A standard metric in object detection representing the area under the precision-recall curve.
- F1 Score:** Combines precision and recall for a balanced accuracy score.

During the training phase, the model learns to identify patterns and features associated with different types of waste through iterative optimization. The training process involves feeding labeled images into the model and updating weights using backpropagation to minimize prediction error. Loss functions such as classification loss, localization loss, and confidence loss are monitored to evaluate training progress. The use of transfer learning significantly speeds up convergence by utilizing pre-trained knowledge. Validation is performed after each epoch to assess model performance on unseen data, ensuring that the model does not overfit. Evaluation metrics such as precision, recall, F1 score, and mean average precision (mAP) are calculated to measure detection accuracy. Confusion matrices and performance graphs are generated to visualize model behavior and identify areas for improvement. Early stopping techniques are applied to prevent unnecessary training once optimal performance is achieved. This structured training and evaluation process ensures that the model is accurate, reliable, and efficient for real-time applications.

5.2.4 MODLE SIMULATION MODULE

The model simulation module forms the core execution layer of the system, where the trained YOLOv8 model and Google Gemini Vision API are activated to process real-time waste images within a fully software-based environment. When a frame is captured through the Streamlit interface, it is passed simultaneously to both models in a parallel inference pipeline. YOLOv8

processes the frame to generate bounding boxes, object labels, and confidence scores, while the Gemini Vision API receives the same frame along with the YOLOv8 output to perform semantic classification, recyclability assessment, contamination analysis, and pixel-level segmentation mask generation. This parallel execution eliminates sequential latency, achieving an average inference time of approximately 1.6 seconds per image. The simulation operates entirely within the local Python runtime, requiring no physical hardware or embedded sensors, making it fully reproducible across different operating systems and development environments. The model simulation module is designed to replicate real-world system behavior within a controlled software environment. It allows users to test the functionality of the waste classification system without requiring physical hardware components. The simulation operates by capturing live input from a camera or processing uploaded images through the Streamlit interface. The parallel inference mechanism ensures that both YOLOv8 and Gemini Vision API process the input simultaneously, reducing overall latency. The module generates outputs such as bounding boxes, class labels, segmentation masks, and recyclability information in real time. It also provides visual feedback to the user, enhancing system usability and transparency. Error handling mechanisms are implemented to manage invalid inputs or processing failures. The simulation environment is optimized for performance, ensuring smooth execution across different systems. This module plays a crucial role in validating system functionality before deployment in real-world scenarios.

5.2.5 USER INTERFACE MODULE

The user interface module is developed using Streamlit, providing an interactive and responsive web-based dashboard for real-time waste analysis. Users can initiate the live camera feed directly through the interface, capture frames for analysis, and receive instant classification results without requiring any technical knowledge. The dashboard displays the captured image alongside the comprehensive waste analysis panel, which includes the object type, material composition, recyclability status indicated by a green or red status banner, condition description, environmental impact information, recycling instructions, and segmentation results. The interface is launched locally using the command `streamlit run app.py` and requires no external server or cloud deployment during simulation. Its clean, modular layout ensures that all outputs from both the YOLOv8 and Gemini pipelines are presented in a structured and user-friendly format, making the system accessible for municipal operators, researchers, and general users alike. The user interface module is designed to provide a seamless and interactive experience for users interacting with the system. Built using Streamlit, the interface enables users to easily upload images or access live camera feeds for waste classification. The layout is structured to display input images alongside

classification results, ensuring clarity and ease of understanding. Visual elements such as bounding boxes, labels, and segmentation masks are overlaid on the image to highlight detected waste objects. Additional information such as recyclability status, material composition, and disposal recommendations is presented in a clear and organized format. The interface is designed to be responsive and accessible across different devices, making it suitable for a wide range of users. User-friendly controls and minimal input requirements ensure that even non-technical users can operate the system efficiently. The modular design of the interface allows for easy updates and feature additions. Overall, the user interface enhances usability, making the system practical for real-world applications.

5.3 SYSTEM TESTING

System testing for the Intelligent Waste Management System is performed to ensure that the complete integrated system meets specified requirements and functions correctly in real-world scenarios, validating the seamless end-to-end interaction between modules like smart sensor data collection, cloud processing, route optimization algorithms, and the user interface. This phase involves evaluating the system for performance (acceptable route calculation time), reliability (consistent operation without crashes), and accuracy (precise fill-level detection and minimal false alerts), often through scenario-based testing like simulating multiple bins reaching capacity or network failures. Ultimately, system testing thoroughly validates the overall functionality, stability, and efficiency of the waste management solution

5.3.1 UNIT TESTING

Waste Management System is a critical, low-level testing phase focused on isolating and verifying the smallest testable parts of the software, such as individual functions, methods, or classes, to ensure they operate correctly according to their design specifications. For the waste management project, this involves rigorously testing core computational and logical components: for example, the sensor data parsing function is tested to guarantee it accurately converts raw electrical signals into reliable fill-level percentages, the route calculation sub-routines are validated to ensure distance calculations between two points are mathematically correct, and the database interaction layer's methods are verified for correct data retrieval and insertion. Crucially, unit tests utilize mock objects and stubs to simulate external dependencies like hardware inputs, network services, or database connections, allowing the component to be tested in complete.

5.3.2 INTEGRATION TESTING

Integration testing was performed to verify the proper interaction and data flow between the different modules of the crash detection system. Unlike unit testing, which focuses on individual components, integration testing ensured that the SensorService, CameraService, MLService, Fusion Engine, Decision Layer, and Emergency Alert Module worked together seamlessly as a complete system. The main objective was to identify interface issues, data mismatches, or failures in communication when modules were combined. Test scenarios included real-time accelerometer data being passed to the machine learning model for crash prediction, camera and EEG/BCI inputs being fused with sensor data, and emergency alerts being triggered based on decisions generated by the system. Special attention was given to timing and synchronization, as crash events require immediate responses within seconds. The testing results demonstrated that modules communicated effectively, data was transferred without loss, and alerts were generated promptly when critical events occurred. This confirmed that the integrated system is capable of functioning reliably in real-world conditions and can respond efficiently during emergencies.

5.3.3 PERFORMANCE TESTING

Performance testing was carried out to evaluate the efficiency, scalability, and responsiveness of the Smart Waste Management system. The goal was to ensure that the machine learning model and web application could handle multiple image uploads and classification requests simultaneously without performance degradation. Tests were conducted to measure the model's inference time, response latency, and accuracy under varying loads. The Streamlit interface and backend model integration were monitored for speed and stability during real-time image classification. Additionally, stress tests were performed to assess how the application behaves under high user traffic. The results showed that the system maintained consistent performance with minimal lag, proving its ability to perform real-time waste classification efficiently. Overall, the performance testing validated that the system can operate reliably in practical environments while maintaining high processing speed and prediction accuracy.

5.3.4 TESTING STRATEGY

The testing strategy adopted for the project followed a systematic and multi-layered approach to ensure complete software reliability. Initially, unit testing was conducted to verify individual components such as data preprocessing, model loading, and classification functions. This was followed by integration testing, which ensured that the connection between the trained model and the Streamlit interface functioned seamlessly. Functional testing validated that the system met all user

requirements, such as accurate image classification and real-time response. Validation and verification were also carried out to confirm that the outputs matched the expected results from the trained dataset. Additionally, cross-browser testing was performed to ensure that the web interface remained responsive across different devices and platforms. This layered testing approach helped identify and eliminate bugs early, ensuring a stable, scalable, and error-free software product.

5.3.5 TOOLS AND FRAMEWORKS

Several tools and frameworks were used to develop, test, and deploy the Smart Waste Management system efficiently. Python served as the core programming language due to its vast ecosystem for machine learning and web development. TensorFlow and Keras were utilized for building, training, and evaluating the CNN model. The user interface was developed using Streamlit, which provided an interactive and lightweight front-end solution. NumPy, Pandas, and OpenCV supported data handling and image preprocessing tasks, while Matplotlib and Seaborn were used for visualization of model performance. For testing and debugging, PyTest and unittest frameworks were used to validate code functionality. Git was employed for version control, and Jupyter Notebook facilitated model experimentation and documentation. Together, these tools provided a robust environment for efficient model development, testing, and deployment.

5.3.6 RESULTS

The Smart Waste Management system successfully achieved its goal of automating waste classification using machine learning. The trained CNN model demonstrated high accuracy in categorizing waste images into biodegradable, non-biodegradable, and recyclable types. During testing, the model achieved consistent performance across multiple datasets, with an accuracy rate exceeding 90% in several test scenarios. The integration with the Streamlit web interface enabled real-time classification, allowing users to upload waste images and instantly receive predictions. Overall, the results validate that the developed software is efficient, accurate, and ready for integration

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The Intelligent Hybrid Waste Classification and Segmentation System developed in this project successfully demonstrates the potential of integrating advanced artificial intelligence techniques for improving waste management practices. By combining the strengths of YOLOv8 for real-time object detection and the Google Gemini Vision API for semantic analysis, the system achieves both speed and depth in classification. This hybrid approach overcomes the limitations of traditional machine learning models that often focus only on surface-level classification without contextual understanding.

The system effectively classifies waste into multiple categories such as plastic, paper, metal, glass, organic, and mixed waste, while also providing additional insights including recyclability status, contamination level, and disposal recommendations. The use of a parallel inference pipeline significantly reduces processing time, enabling near real-time performance suitable for practical applications. Furthermore, the integration of Explainable AI techniques using SHAP enhances transparency by identifying key features that influence model predictions, addressing one of the major challenges associated with black-box AI systems.

The implementation of the system through a Streamlit-based interface ensures ease of use and accessibility, allowing users to interact with the model without requiring technical expertise. The simulation-based approach adopted in this project enables thorough testing and validation of the system before hardware integration, ensuring reliability and scalability. The results obtained indicate that the system performs with high accuracy and consistency across different test scenarios, making it a viable solution for real-world deployment.

Overall, this project establishes a strong foundation for intelligent waste management by demonstrating how AI-driven solutions can automate waste classification, reduce manual effort, and contribute to environmental sustainability. The modular and scalable architecture further ensures that the system can be extended and adapted for future technological advancements, positioning it as a practical solution aligned with smart city initiatives and sustainable development goals, addressing one of the major challenges associated with black-box AI systems.

6.2 FUTURE WORK

While the current system provides a robust software-based solution for waste classification, there are several opportunities for future enhancements that can further improve its functionality and real-world applicability. One of the primary areas of future work involves the integration of IoT-enabled smart bins equipped with sensors such as ultrasonic and weight sensors. This would enable automated waste level detection and real-time monitoring, allowing for more efficient waste collection and management.

Another important extension is the deployment of the system on edge devices such as Raspberry Pi or embedded systems, enabling on-device processing without reliance on cloud infrastructure. This would reduce latency, improve response time, and make the system suitable for use in remote or low-connectivity areas. Additionally, optimizing the machine learning model to run efficiently on low-power devices will be a key focus for future development.

The system can also be enhanced by integrating cloud-based platforms for centralized data storage and analytics. This would enable large-scale data collection, trend analysis, and predictive insights for better decision-making in municipal waste management systems. A dedicated mobile application can be developed to provide users with easy access to the system, allowing real-time interaction and monitoring through smartphones.

Further improvements may include expanding the dataset to cover a wider range of waste categories and environmental conditions, thereby improving model accuracy and generalization. Advanced deep learning techniques and newer model architectures can be explored to further enhance detection performance. Integration with recycling units and waste-to-energy systems can also be considered to create a complete end-to-end waste management ecosystem.

In addition, the system can be extended to include features such as route optimization for waste collection vehicles using AI algorithms, helping reduce fuel consumption and operational costs. The incorporation of multilingual support and voice-based interaction can improve accessibility for diverse user groups. Overall, future developments aim to transform the current system into a fully autonomous, intelligent, and scalable waste management solution capable of addressing real-world challenges in urban environments.

CHAPTER 7

APPENDIX

7.1 SOURCE CODE

Dataset Module:

```
import os
import json
import shutil
from pathlib import Path
from PIL import Image
from pycocotools.coco import COCO

ROOT = Path("dataset")
TACO_DIR = ROOT / "TACO"
TRASHNET_DIR = ROOT / "TrashNet"
YOLO_DIR = ROOT / "yolo_dataset"
CLASSES = ["plastic", "metal", "paper"]

def ensure_dirs():
    (YOLO_DIR / "images" / "train").mkdir(parents=True, exist_ok=True)
    (YOLO_DIR / "images" / "val").mkdir(parents=True, exist_ok=True)
    (YOLO_DIR / "labels" / "train").mkdir(parents=True, exist_ok=True)
    (YOLO_DIR / "labels" / "val").mkdir(parents=True, exist_ok=True)

def copy_trashnet():
    imgs = list((TRASHNET_DIR/"images").glob("*"))
    split = int(0.8*len(imgs))
    for i,f in enumerate(imgs):
        dest_img = YOLO_DIR/"images"/("train" if i<split else "val")/f.name
        shutil.copy(f, dest_img)
        label_path = YOLO_DIR/"labels"/("train" if i<split else "val")/(f.stem+".txt")
        cls = 0
        w,h = Image.open(f).size
        with open(label_path,"w") as L:
```

```
L.write(f'{cls} 0.5 0.5 0.8 0.8\n')
```

```
def coco_to_yolo(coco_ann_path, image_src_dir, dest_split="train"):
    coco = COCO(coco_ann_path)
    img_ids = coco.getImgIds()
    for idx, img_id in enumerate(img_ids):
        img_info = coco.loadImgs(img_id)[0]
        fname = img_info["file_name"]
        src = Path(image_src_dir)/fname
        if not src.exists():
            continue
        dest_img = YOLO_DIR/"images"/dest_split/fname
        shutil.copy(src, dest_img)
        ann_ids = coco.getAnnIds(imgIds=img_id)
        anns = coco.loadAnns(ann_ids)
        label_file = YOLO_DIR/"labels"/dest_split/(Path(fname).stem+".txt")
        with open(label_file, "w") as lf:
            for a in anns:
                cat = coco.loadCats(a["category_id"])[0]["name"]
                if cat.lower() in CLASSES:
                    cls = CLASSES.index(cat.lower())
                else:
                    if "plastic" in cat.lower():
                        cls = 0
                    elif "metal" in cat.lower():
                        cls = 1
                    else:
                        cls = 2
                x, y, w, h = a["bbox"]
                img_w, img_h = img_info["width"], img_info["height"]
                cx = (x + w/2)/img_w
                cy = (y + h/2)/img_h
                nw = w/img_w
                nh = h/img_h
```

```

        lf.write(f'{cls} {cx:.6f} {cy:.6f} {nw:.6f} {nh:.6f}\n')

def create_data_yaml():
    data = {
        "train": str((YOLO_DIR/"images"/"train").resolve()),
        "val": str((YOLO_DIR/"images"/"val").resolve()),
        "nc": len(CLASSES),
        "names": CLASSES
    }
    with open(YOLO_DIR/"data.yaml", "w") as f:
        json.dump(data, f)

def main():
    ensure_dirs()
    copy_trashnet()
    coco_to_yolo(TACO_DIR/"annotations.json", TACO_DIR/"images", "train")
    create_data_yaml()

if __name__ == "__main__":
    main()

```

Model Module

```

from ultralytics import YOLO
import yaml
model = YOLO("yolov8n.pt")
results = model.train(data="dataset/yolo_dataset/data.yaml", epochs=50, imgsz=640, batch=16,
name="trashnet_yolov8n")
model = YOLO("runs/detect/trashnet_yolov8n/weights/best.pt")
model.export(format="onnx", imgsz=640)
model.export(format="saved_model", imgsz=640)

```

Detection Module

```

const video = document.getElementById("video");

```

```

const canvas = document.getElementById("overlay");
const ctx = canvas.getContext("2d");
let model = null;
async function setupCamera(){
  const stream = await
navigator.mediaDevices.getUserMedia({ video:{facingMode:"environment"}, audio:false});
  video.srcObject = stream;
  await video.play();
  canvas.width = video.videoWidth;
  canvas.height = video.videoHeight;
}
function preprocess(input){
  const tfImg = tf.browser.fromPixels(input).toFloat();
  const resized = tf.image.resizeBilinear(tfImg, [640,640]);
  const normalized = resized.div(255.0);
  const expanded = normalized.expandDims(0);
  return expanded;
}
function drawBoxes(bboxes, scores, classes){
  ctx.clearRect(0,0,canvas.width,canvas.height);
  for(let i=0;i<bboxes.length;i++){
    const b = bboxes[i];
    const x = b[0]*canvas.width;
    const y = b[1]*canvas.height;
    const w = (b[2]-b[0])*canvas.width;
    const h = (b[3]-b[1])*canvas.height;
    ctx.strokeStyle = "lime";
    ctx.lineWidth = 2;
    ctx.strokeRect(x,y,w,h);
    ctx.fillStyle = "lime";
    ctx.font = "18px Arial";
    ctx.fillText(` ${classes[i]} ${((scores[i]*100).toFixed(1))}%`, x, y>20? y-5: y+20);
  }
}

```

```

async function runDetection(){
  const inputTensor = preprocess(video);
  const preds = await model.executeAsync(inputTensor);
  const boxes = preds[0].arraySync();
  const scores = preds[1].arraySync();
  const classes = preds[2].arraySync();
  const flatBoxes = [];
  const flatScores = [];
  const flatClasses = [];
  for(let i=0;i<boxes.length;i++){
    flatBoxes.push([boxes[i][1], boxes[i][0], boxes[i][3], boxes[i][2]]);
    flatScores.push(scores[i]);
    flatClasses.push(["plastic","metal","paper"][classes[i]]);
  }
  const nmsIndices = await tf.image.nonMaxSuppressionAsync(tf.tensor2d(flatBoxes),
tf.tensor1d(flatScores), 20, 0.5, 0.4);
  const idx = await nmsIndices.array();
  const selBoxes = idx.map(i=>flatBoxes[i]);
  const selScores = idx.map(i=>flatScores[i]);
  const selClasses = idx.map(i=>flatClasses[i]);
  drawBoxes(selBoxes, selScores, selClasses);
  tf.dispose([inputTensor, preds, nmsIndices]);
  requestAnimationFrame(runDetection);
}

async function init(){
  await setupCamera();
  model = await tf.loadGraphModel("tfjs_model/model.json");
  requestAnimationFrame(runDetection);
}

window.addEventListener("load", init);

```

app.js

```
import streamlit as st
```

```

import cv2

from PIL import Image as PILImage

import concurrent.futures

import os

import time

from yolo_detect import detect_objects


# Import functions from existing files

import importlib.util


# Import object-detection.py (handle hyphenated filename)

spec = importlib.util.spec_from_file_location("object_detection", "object-detection.py")

object_detection = importlib.util.module_from_spec(spec)

spec.loader.exec_module(object_detection)

extract_segmentation_masks = object_detection.extract_segmentation_masks


# Page configuration

st.set_page_config(

    page_title="Waste Classification & Segmentation",

    page_icon="🗑️",

    layout="wide",

    initial_sidebar_state="expanded"

)


# Custom CSS for cleaner UI

st.markdown("""

```

```

<style>

.main-header {

    text-align: center;

    padding: 1rem 0;

    border-bottom: 2px solid #1f77b4;

    margin-bottom: 2rem;

}

.stButton>button {

    width: 100%;

    border-radius: 8px;

    height: 3rem;

    font-weight: 600;

}

.result-box {

    padding: 1rem;

    border-radius: 8px;

    background-color: #f0f2f6;

    margin: 1rem 0;

}

</style>

""", unsafe_allow_html=True)

```

```

st.markdown('<div      class="main-header"><h1> Waste Classification & Segmentation</h1><p>Real-time AI-powered waste analysis</p></div>', unsafe_allow_html=True)

```

```

# Initialize session state

```

```

if 'captured_image' not in st.session_state:
    st.session_state.captured_image = None

if 'gemini_response' not in st.session_state:
    st.session_state.gemini_response = None

if 'segmentation_results' not in st.session_state:
    st.session_state.segmentation_results = None

if 'processing' not in st.session_state:
    st.session_state.processing = False

if 'live_feed_active' not in st.session_state:
    st.session_state.live_feed_active = False

if 'stop_feed' not in st.session_state:
    st.session_state.stop_feed = False

if 'capture_triggered' not in st.session_state:
    st.session_state.capture_triggered = False


# Sidebar for camera settings
with st.sidebar:
    st.header("⚙️ Settings")

    camera_index = st.selectbox("📷 Camera Index", [0, 1, 2], index=1)

    output_dir = st.text_input("📁 Output Directory", value="segmentation_outputs")

    st.markdown("---")


# Status indicators
st.header("📊 Status")

if st.session_state.live_feed_active:
    st.success("✅ Live Feed Active")

```

else:

st.info("○ Feed Inactive")

if st.session_state.processing:

st.warning("⌛ Processing...")

elif st.session_state.gemini_response:

st.success("✅ Analysis Complete")

st.markdown("---")

if st.button("🗑️ Clear All Results", width='stretch'):

st.session_state.gemini_response = None

st.session_state.segmentation_results = None

st.session_state.captured_image = None

st.session_state.live_feed_active = False

st.session_state.stop_feed = True

st.session_state.capture_triggered = False

st.rerun()

Main content area

col1, col2 = st.columns([1.2, 1])

with col1:

st.header("📹 Live Camera Feed")

Camera controls

button_col1, button_col2, button_col3 = st.columns([1, 1, 1])

```

with button_col1:

start_feed = st.button("▶ Start Live Feed", width='stretch', type="primary")

if start_feed:

st.session_state.live_feed_active = True

st.session_state.stop_feed = False

st.session_state.capture_triggered = False


with button_col2:

stop_feed = st.button("⏏ Stop Feed", width='stretch')

if stop_feed:

st.session_state.live_feed_active = False

st.session_state.stop_feed = True


with button_col3:

capture_button = st.button("📷 Capture", width='stretch',

                             disabled=not st.session_state.live_feed_active)

if capture_button and st.session_state.live_feed_active:

    st.session_state.capture_triggered = True

    st.session_state.live_feed_active = False

    st.session_state.stop_feed = True


# Live feed placeholder

live_feed_placeholder = st.empty()


# Display live feed using auto-refresh

```

```

if st.session_state.live_feed_active and not st.session_state.stop_feed:

    cap = cv2.VideoCapture(camera_index)

    if not cap.isOpened():

        with live_feed_placeholder.container():

            st.error(f"✘ Cannot open camera {camera_index}. Try a different camera index.")

            st.session_state.live_feed_active = False

    else:

        ret, frame = cap.read()

        if ret:

            # Convert BGR to RGB

            frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)

            live_feed_placeholder.image(frame_rgb, caption="Live Camera Feed", channels="RGB")

            # Auto-refresh to show live feed

            time.sleep(0.05) # Small delay

            st.rerun()

        else:

            with live_feed_placeholder.container():

                st.error("✘ Failed to read from camera")

                st.session_state.live_feed_active = False

            cap.release()

elif st.session_state.capture_triggered and not st.session_state.captured_image:

    # Capture frame when button is pressed

    cap = cv2.VideoCapture(camera_index)

    if cap.isOpened():

        ret, frame = cap.read()

```

if ret:

```
frame_rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
```

```
temp_image_path = "temp_captured_image.jpg"
```

```
cv2.imwrite(temp_image_path, frame)
```

```
st.session_state.captured_image = {
```

```
    'path': temp_image_path,
```

```
    'image': frame_rgb
```

```
}
```

```
st.session_state.capture_triggered = False
```

```
st.success("✅ Image captured successfully!")
```

```
cap.release()
```

```
cv2.destroyAllWindows()
```

else:

```
with live_feed_placeholder.container():
```

```
    if not st.session_state.live_feed_active:
```

```
        st.info("👉 Click 'Start Live Feed' to begin")
```

Show captured image if available

```
if st.session_state.captured_image:
```

```
    st.markdown("---")
```

```
    st.subheader("📷 Captured Image")
```

```
    st.image(st.session_state.captured_image['image'], caption="Ready to Process",
channels="RGB")
```

Process button

```
if st.button("🔍 Analyze Image", width='stretch', type="primary",
```

```

        disabled=st.session_state.processing):

st.session_state.processing = True

st.session_state.capture_triggered = False


# Capture image path before threading
image_path = st.session_state.captured_image['path']


# ----- YOLO DETECTION (NEW) -----

detections = detect_objects(image_path)


# Convert YOLO output into text context for Gemini
yolo_context = ""

if detections:

    for d in detections:

        yolo_context += (

            f"Detected object: {d['label']} "

            f"with confidence {d['confidence']:.2f}.\n"

        )

    else:

        yolo_context = "No objects were confidently detected by YOLO."

# -----


# Create placeholders for results

progress_placeholder = st.empty()


with progress_placeholder.container():

```

```
st.info("🔄 Processing image with Gemini AI and generating segmentation masks...")
```

```
# Run both processes in parallel
```

```
def run_gemini_classification(img_path, yolo_context):
```

```
    try:
```

```
        from google import genai
```

```
        from google.genai.errors import ServerError, APIError
```

```
        import time
```

```
        client = genai.Client(api_key="AIzaSyAgvzUpebcc-UFxcKjwWcdc7sjGNdVMvP8")
```

```
        with open(img_path, "rb") as img_file:
```

```
            image_bytes = img_file.read()
```

```
        prompt_text = (
```

```
            "You are an expert waste management and recycling specialist.\n\n"
```

```
            "Preliminary object detection using YOLO has identified the following:\n"
```

```
            f"{yolo_context}\n\n"
```

```
            "Using the above detection information AND the image, analyze the waste and "
```

```
            "provide a comprehensive classification.\n\n"
```

```
            "Please provide your response in the following structured format:\n"
```

```
            "STATUS:\nOBJECT:\nMATERIAL:\nCONDITION:\n"
```

```
            "RECYCLING_INSTRUCTIONS:\nALTERNATIVES:\n"
```

```
            "ENVIRONMENTAL_IMPACT:\nSPECIAL_NOTES:\n"
```

```
        )
```

```
        max_retries = 5
```

```
        for attempt in range(1, max_retries + 1):
```

```
            try:
```

```
                response = client.models.generate_content(
```

```

        model="gemini-2.5-flash",
        contents=[
            {
                "role": "user",
                "parts": [
                    {"text": prompt_text},
                    {
                        "inline_data": {
                            "mime_type": "image/jpeg",
                            "data": image_bytes
                        }
                    }
                ]
            }
        ]
    )

    return response.text or " ✖ Gemini returned an empty response"

except (ServerError, APIError) as e:
    if attempt < max_retries:
        time.sleep(2 ** attempt)
    else:
        return f" ✖ Gemini error after retries: {str(e)}"

except Exception as e:
    return f" ✖ Gemini wrapper error: {str(e)}"

except Exception as e:
    return f" ✖ Gemini initialization error: {str(e)}"

```

```

def run_segmentation(img_path, out_dir):
    """Run segmentation and return results"""
    try:
        combined_overlay_path = extract_segmentation_masks(img_path, out_dir)
        return {
            'success': True,
            'combined_overlay_path': combined_overlay_path,
            'output_dir': out_dir,
            'has_masks': combined_overlay_path is not None
        }
    except Exception as e:
        return {
            'success': False,
            'error': str(e),
            'has_masks': False
        }

# Execute both processes in parallel
with concurrent.futures.ThreadPoolExecutor(max_workers=2) as executor:
    gemini_future = executor.submit(
        run_gemini_classification,
        image_path,
        yolo_context
    )
    segmentation_future = executor.submit(run_segmentation, image_path, output_dir)

```

```

        # Wait for both to complete

        gemini_result = gemini_future.result()

        segmentation_result = segmentation_future.result()


    # Update session state

    st.session_state.gemini_response = gemini_result

    st.session_state.segmentation_results = segmentation_result

    st.session_state.processing = False


    # Clear progress and show results

    progress_placeholder.empty()


    # Display results in right column will be updated automatically


with col2:

    st.header("📊 Analysis Results")


    # Display Gemini Response

    if st.session_state.gemini_response:

        with st.container():

            st.markdown("### 🔍 Comprehensive Waste Analysis")

            st.markdown("---")


            response_text = st.session_state.gemini_response

```

```

# Parse structured response

def parse_response(text):
    """Parse the structured Gemini response"""

    parsed = {}

    lines = text.split('\n')

    current_key = None
    current_value = []

    for line in lines:

        line = line.strip()

        if ':' in line and not line.startswith(' '):

            # Save previous key-value

            if current_key:

                parsed[current_key] = '\n'.join(current_value).strip()

            # Extract new key-value

            parts = line.split(':', 1)

            if len(parts) == 2:

                current_key = parts[0].strip()

                current_value = [parts[1].strip()] if parts[1].strip() else []

            else:

                current_key = None

                current_value = []

        elif current_key and line:

            current_value.append(line)

        elif not current_key and line:

```

```

# Free-form text at the beginning

if 'FREE_FORM' not in parsed:

    parsed['FREE_FORM'] = []

    parsed['FREE_FORM'].append(line)


# Save last key-value

if current_key:

    parsed[current_key] = '\n'.join(current_value).strip()


return parsed


parsed_response = parse_response(response_text)


# Display Status

status = parsed_response.get('STATUS', '')

if status:

    status_lower = status.lower()

    if 'recyclable' in status_lower and 'non' not in status_lower and 'partially' not in
status_lower:

        st.success(f"✅ **Status: {status}**")

    elif 'partially' in status_lower:

        st.warning(f"⚠️ **Status: {status}**")

    elif 'non-recyclable' in status_lower:

        st.error(f"❌ **Status: {status}**")

    else:

        st.info(f"📄 **Status: {status}**")

```

```

# Display Object and Material in columns

col_obj, col_mat = st.columns(2)

with col_obj:

    if parsed_response.get('OBJECT'):

        st.metric("Object", parsed_response['OBJECT'])

with col_mat:

    if parsed_response.get('MATERIAL'):

        st.metric("Material", parsed_response['MATERIAL'])


# Display Condition

if parsed_response.get('CONDITION'):

    condition = parsed_response['CONDITION']

    condition_lower = condition.lower()

    if 'good' in condition_lower or 'usable' in condition_lower:

        st.info(f"🟢 **Condition:** {condition}")

    elif 'damaged' in condition_lower or 'contaminated' in condition_lower:

        st.warning(f"🔴 **Condition:** {condition}")

    else:

        st.caption(f"***Condition:** {condition}")


# Display Recycling Instructions

if parsed_response.get('RECYCLING_INSTRUCTIONS'):

    with st.expander("♻️ Recycling Instructions", expanded=True):

        instructions = parsed_response['RECYCLING_INSTRUCTIONS']

        # Format as bullet points if it's a list

```

```

    if '\n' in instructions or '•' in instructions or '-' in instructions:

        st.markdown(instructions)

    else:

        st.markdown(f"• {instructions}")


# Display Alternatives

if parsed_response.get('ALTERNATIVES'):

    with st.expander("🔗 Alternatives & Reuse Options"):

        st.markdown(parsed_response['ALTERNATIVES'])


# Display Environmental Impact

if parsed_response.get('ENVIRONMENTAL_IMPACT'):

    with st.expander("🌍 Environmental Impact"):

        st.info(parsed_response['ENVIRONMENTAL_IMPACT'])


# Display Special Notes

if parsed_response.get('SPECIAL_NOTES'):

    with st.expander("📝 Special Notes"):

        st.markdown(parsed_response['SPECIAL_NOTES'])


# If response doesn't follow structure, show full response

if not parsed_response or len(parsed_response) <= 1:

    with st.expander("📖 Full Analysis"):

        st.text_area("", value=response_text, height=200, disabled=True,
label_visibility="collapsed")

```

```

# Display Segmentation Results

if st.session_state.segmentation_results:

    st.markdown("### 🤖 Segmentation Results")

    st.markdown("---")


if st.session_state.segmentation_results['success']:

    if st.session_state.segmentation_results.get('has_masks', False):

        combined_path = st.session_state.segmentation_results.get('combined_overlay_path')

        if combined_path and os.path.exists(combined_path):

            st.success("✅ Masks Generated Successfully!")


# Display combined overlay

overlay_img = PILImage.open(combined_path)

st.image(overlay_img, caption="Segmentation Overlay")


# Count masks

output_dir = st.session_state.segmentation_results.get('output_dir',
'segmentation_outputs')

if os.path.exists(output_dir):

    mask_files = [f for f in os.listdir(output_dir) if f.endswith('_mask.png')]

    if mask_files:

        st.info(f"📁 Detected **{len(mask_files)}** object(s)")

    else:

        st.info("No overlay image available")

else:

    st.warning("⚠️ No objects detected")

```

```
st.caption("The image was processed but no objects were found for segmentation. Try capturing a clearer image with visible objects.")
```

```
else:
```

```
st.error(f"❌ Segmentation failed: {st.session_state.segmentation_results.get('error', 'Unknown error')}")
```

```
else:
```

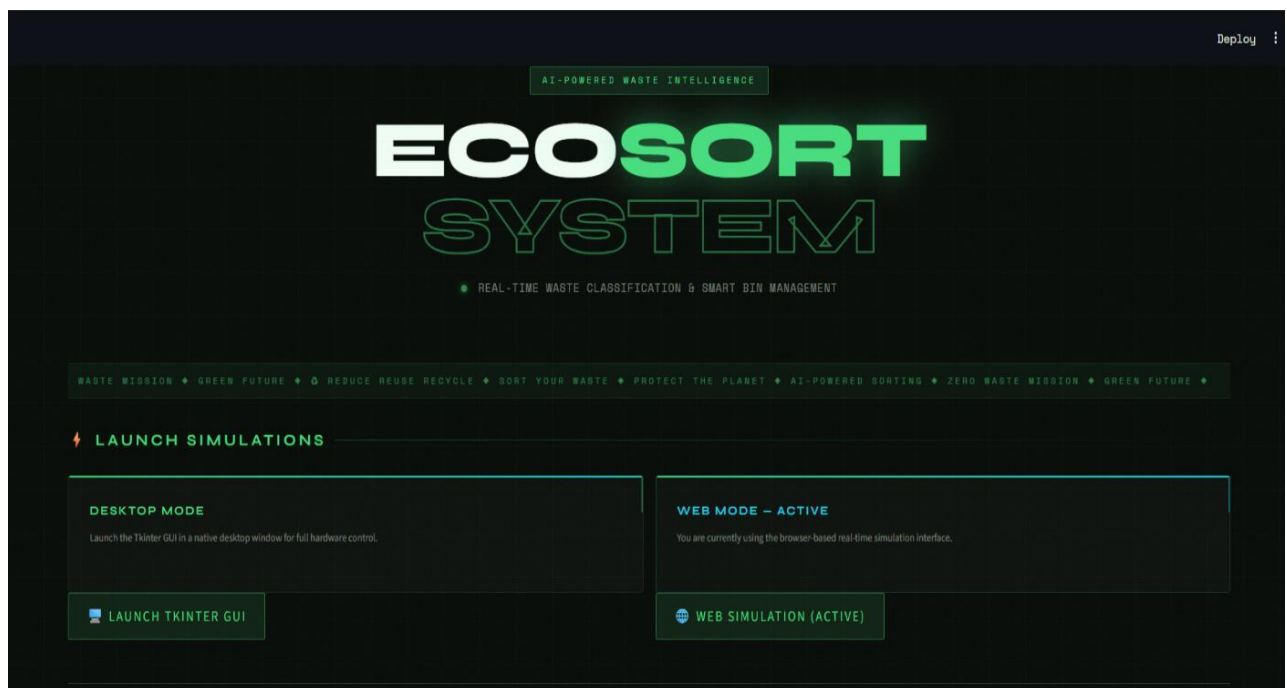
```
st.info("📷 Capture an image and click 'Analyze Image' to see results here")
```

```
# Footer
```

```
st.markdown("----")
```

```
st.caption("💡 **Tip:** Start the live feed, position your object, then capture to analyze it.")
```

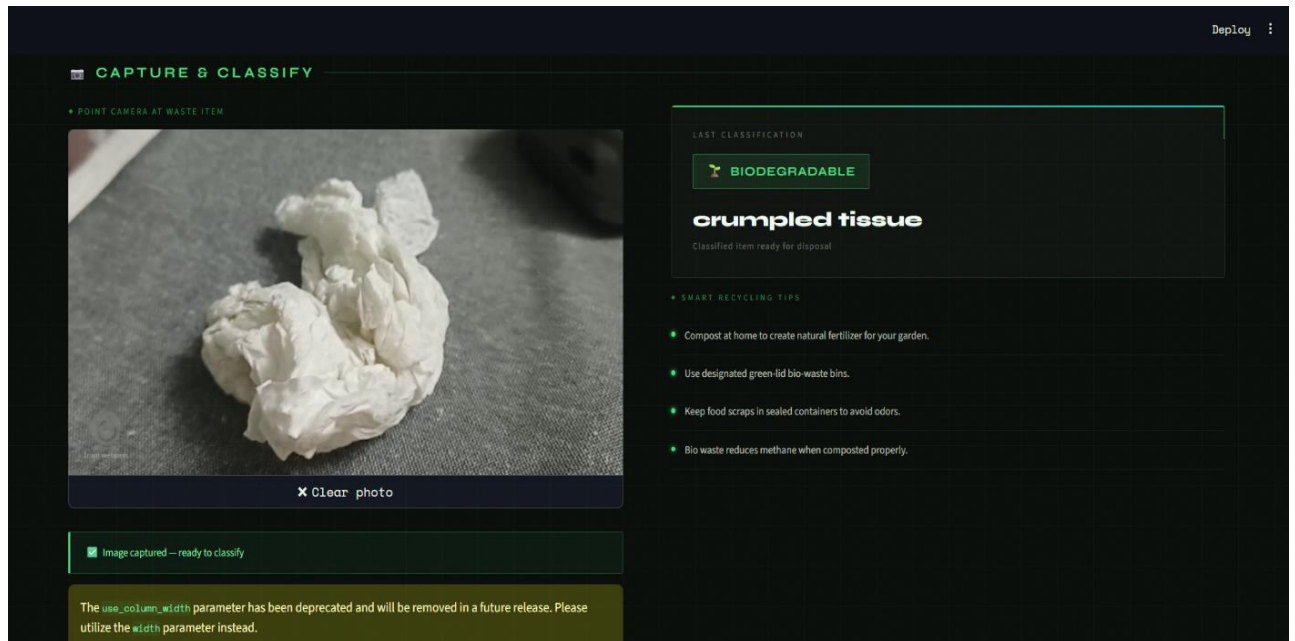
7.2 SCREEN SHOTS



A1. Home Page

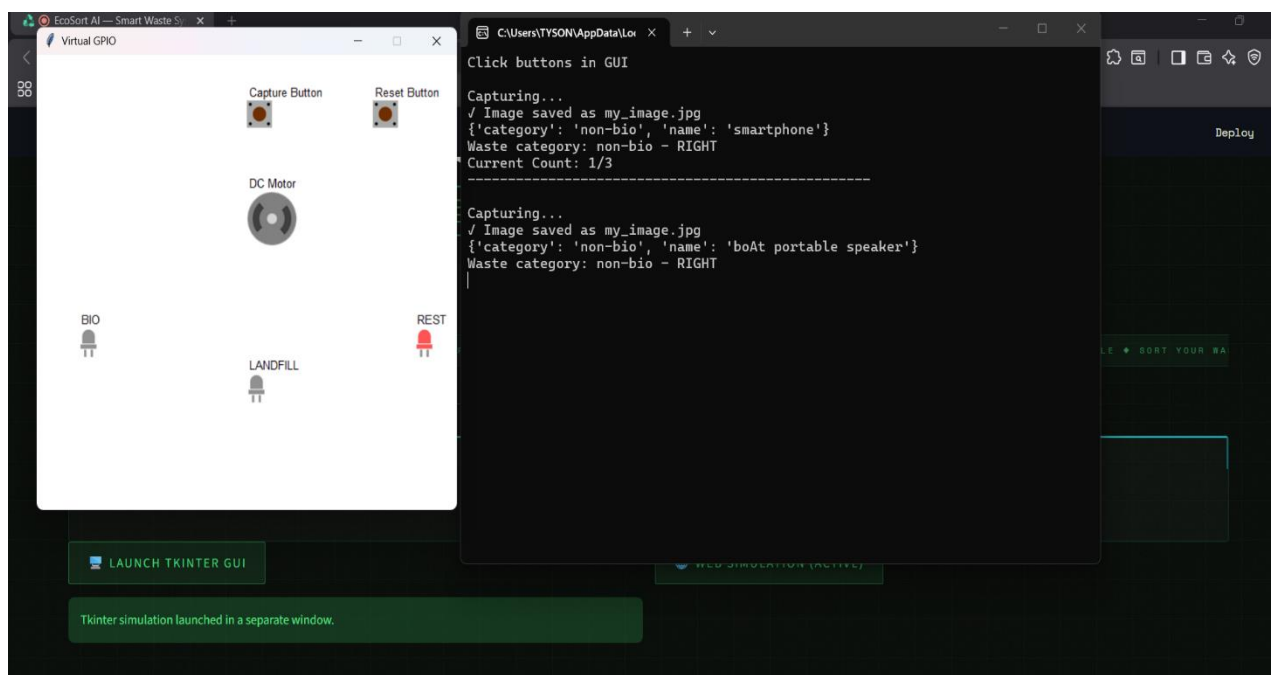
The Home Page serves as the entry point of the Smart Waste Management System. It is developed using Streamlit and provides a clean, user-friendly interface for interacting with the system. The page allows users to start the application, access the live camera feed, or upload images for waste classification. It also displays basic instructions to guide users on how to use the system

effectively. The design focuses on simplicity and accessibility.



A2: Main Output Interface

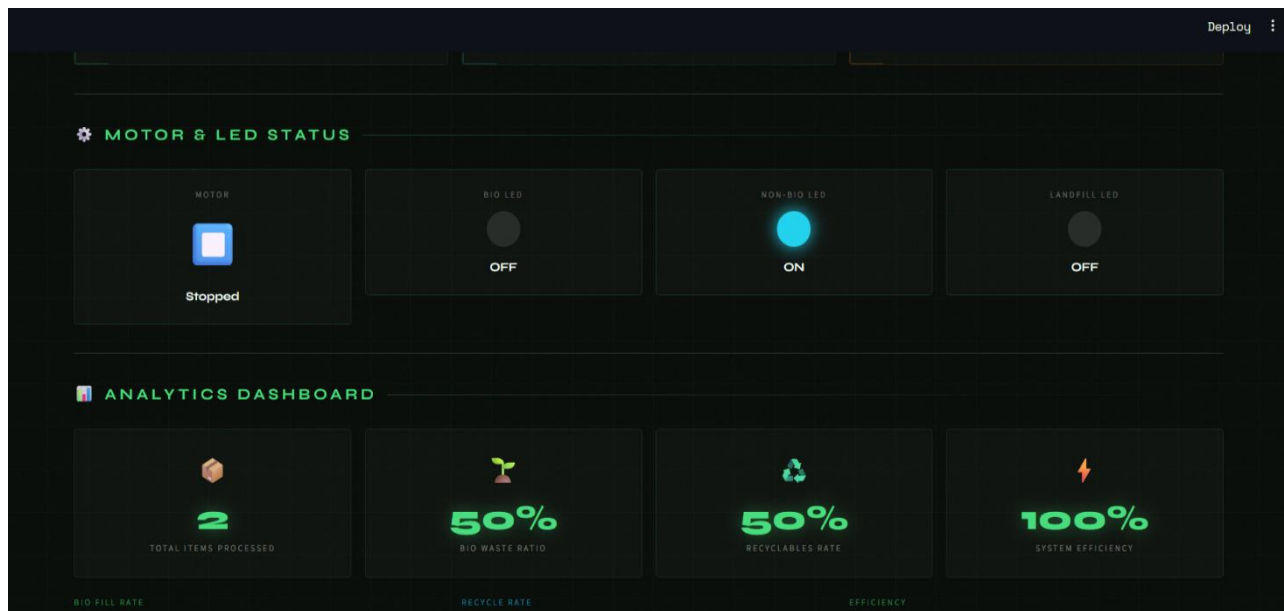
The Main Output Interface displays the results of waste classification in real time. Once an image is captured or uploaded, the system processes it using the YOLOv8 model and Gemini Vision API. The output includes detected objects with bounding boxes, class labels, and confidence scores. Additionally, detailed information such as waste category (plastic, metal, paper, organic, etc.), recyclability status, and disposal suggestions are shown. This interface ensures that users can clearly understand the classification results and take appropriate actions.



A3: Simulation Interface

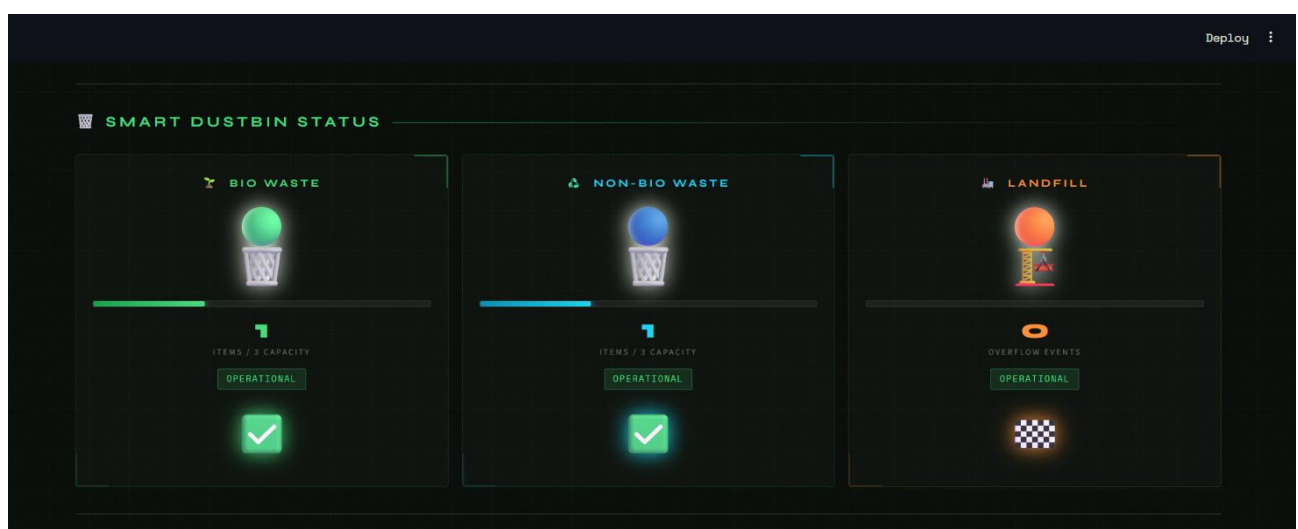
The Simulation Interface represents the real-time execution environment of the system. It allows

users to interact with the live camera feed and observe how the model performs detection and classification dynamically. The system processes each frame and displays results instantly, demonstrating the efficiency of the parallel inference architecture. This interface highlights the system’s capability to function without physical hardware, making it suitable for simulation-based deployment and testing.



A4: Analytics Display

The Analytics Display provides insights into the system’s performance and waste classification trends. It shows graphical representations such as the number of detected waste items, category-wise distribution, and system accuracy metrics. This section helps users and administrators analyze waste patterns and evaluate model performance. The analytics feature supports data-driven decision-making and enhances the overall effectiveness of the waste management system.



A5: Fill Level Indicator

The Fill-Level Indicator visually represents the proportion of different types of waste detected by the system. It uses dynamic bars or indicators to show the level of biodegradable, non-biodegradable, and recyclable waste. This feature simulates the behavior of a smart waste bin by providing real-time status updates. It helps users understand waste accumulation and can be extended in future to trigger alerts when bins reach capacity.

CHAPTER 8

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